POLYMERS 2

2

Plastics and heat

Plastics are divided into two major groups according to their behaviour when exposed to heat. **Thermoplastic plastics** soften when heated and harden on cooling. This means that they can be melted down and reformed many times. **Thermosetting plastics** can be moulded when they are initially liquid but once set, do not soften when heated. Bakelite was the first thermosetting plastic.

1 Give an example of when a *thermosetting plastic* is more useful than a *thermoplastic plastic*.

2 Give an example of when a *thermoplastic plastic* is more useful than a *thermosetting plastic*.

Polyethylene There are two forms of polyethylene (polythene): low-density polyethylene (LDPE) and high-density polyethylene (HDPE). Both polymers have ethylene (ethene) as the monomer. Figure 1 shows the structures of ethylene and polyethylene. The polyethylene molecule consists of thousands of repeating units; the diagram only shows a small part of the chain.

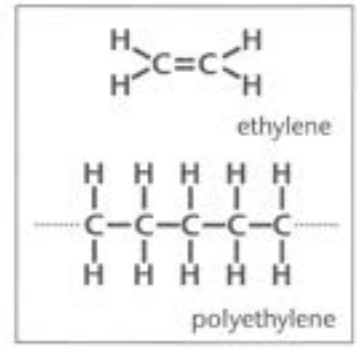


Figure 1

3 Give the molecular formula of ethylene.

4 The molecular formula of polyethylene is often written as $-(C_2H_4)_n-$. Explain what is meant by this formula.

Polyethylene is an **addition polymer**. It is formed by many ethylene molecules joining together when the carbon–carbon double bond breaks, and forms bonds with adjacent molecules. Polyethylene is very cheap, strong, flexible and chemically unreactive. LDPE is used to make packaging materials (especially supermarket shopping bags) and HDPE is used for containers and automotive fittings. Because polyethylene is unreactive, discarded items made of this material don't break down but remain in the environment. If they are collected, they can be recycled. Polythene shopping bags have been handed out by retailers for many years. They are ideal for this purpose, but many people want to reduce their use.

5 Define *polymer*.

6 Polyethylene is a thermoplastic plastic. What is meant by *thermoplastic*?

7 What feature of the ethylene molecule allows it to link with other ethylene molecules?

8 Give two properties of LDPE that make it suitable for making shopping bags.

9 Why do some people want to reduce the numbers of polyethylene bags used?

Polyvinylchloride (PVC) This polymer is also an addition polymer. The monomer is monochloroethylene (vinyl chloride). PVC is strong and weather resistant. It is used for pipes, house cladding and enclosures for electronic equipment. It can also be treated to form a softer substance that is used in shrink wrapping and food packaging.

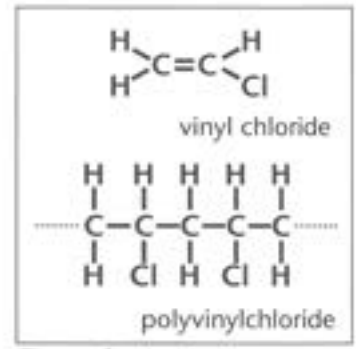


Figure 2

10 Give the molecular formula for vinyl chloride (see figure 2).

11 Give the molecular formula for PVC. (Hint: see question 3.)

Polypropylene This polymer is similar in structure to polyethylene. It is also inexpensive but is stronger than polyethylene. It is used in making bottles, ropes, carpets and car parts. Figure 3 shows a short length of a polypropylene molecule.

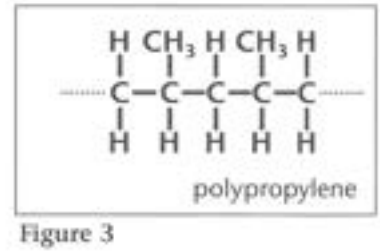


Figure 3

12 Why is only part of the polymer shown?

13 What is the name of the monomer in polypropylene?

14 Draw the structural formula of the monomer in polypropylene.

15 Give the molecular formula of the monomer in polypropylene.

16 Is propylene likely to be *thermoplastic* or *thermosetting*? Explain your answer.

17 Petrochemicals are the raw materials used to make many plastics. What are *petrochemicals*?

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Polystyrene This is a rigid thermoplastic. Like many plastics, it is an excellent electrical insulator and is resistant to attack by chemicals. It is used to make disposable knives and forks, model kits and housings for large household appliances. It can also be 'foamed' to form styrene foam or Styrofoam™ which is much less dense than most materials. In this form it is used to make insulation for food and beverage containers, the 'beans' in 'bean bags' and protective packaging. Figure 4 gives the structure of the monomer in polystyrene: styrene (ethenylbenzene).

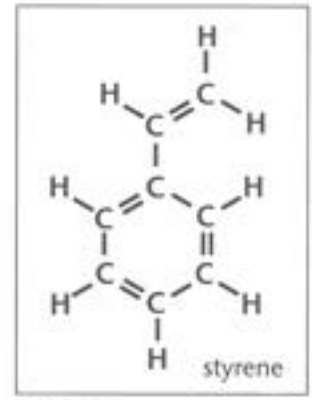


Figure 4

18 Give the molecular formula of styrene.

19 Why is expanded polystyrene a good material for making swimming aids and toys designed to float?

Polytetrafluoroethylene (PTFE) This polymer has the trade names 'TEFAL™' and 'TEFLON™'. The monomer is tetrafluoroethylene and has the formula C_2F_4. PTFE is slippery and tough; it resists most chemicals and can withstand a wide range of temperatures. It is used in gaskets, sealants and protective coatings. The most common domestic use of PTFE is as the 'non-stick' coating on cooking utensils.

20 Draw the structure of tetrafluoroethylene. (Hint: there is a carbon-carbon double bond.)

21 Draw a section of the PTFE molecule.

Synthetic rubber There have been many types of synthetic rubbers produced; 'Buna' rubber was among the first. This polymer is not made up of one repeating monomer (**homopolymer**) but has two different alternating monomers (**copolymer**). Buna rubber has butadiene and styrene as monomers. It is named after *bu*tadiene and sodium, *Na*, which is a catalyst in its formation. World War II caused a shortage of natural rubber, which played a major role in the development of synthetic rubbers. After the war the Germans focused their research on producing a synthetic rubber that was better quality than the existing methyl rubber. Buna rubber was the result of their research.

22 Explain the difference between a *homopolymer* and a *copolymer*.

23 Buna rubber is made using butadiene, the simplest of the dienes. Define *diene*.

Nylon Nylon was originally a tradename but it is now a generic name for a range of polyamides. A polyamide has repeating amide (R–CO–NH–R') groups. R and R' refer to different hydrocarbon groups. The two earliest nylons were Nylon 6 and Nylon 6, 6. Both of these are still very important commercially today. Figure 5 shows the similarity between the structures of these two polymers. Nylonis strong, wear-resistant and can be dyed. It is used to make clothing, carpets, parachutes, ropes and machinery parts. Nylon is a **condensation** polymer. Addition polymers have carbon–carbon double bonds in monomers, which break open and join with neighbouring monomers. Condensation polymers form when an –H 'ending' and an –OH molecule 'ending' from adjacent monomers are removed when the monomers combine. When monomers bond in condensation polymerisation, molecules of water are formed as well as the polymer chain.

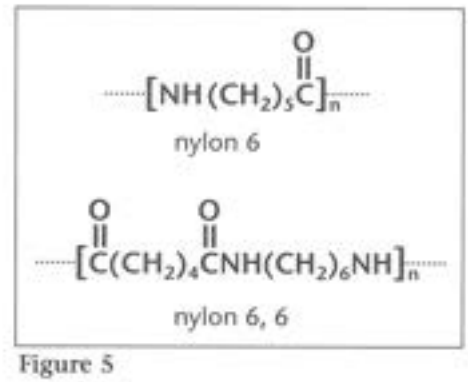


Figure 5

24 What is meant by a *generic* name?

25 Name the elements present in a polyamide.

26 Outline the major differences in bonding between an addition polymer and a condensation polymer.

27 Give one reason for the invention of synthetic rubber.

28 Draw up a table showing the names and uses of the plastics mentioned on these pages.

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